

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Scenario	How the multimeter helps
Laptop does not power on	Measures the laptop charger voltage to check if the adapter is supplying the correct power.
Smartphone not charging	Checks the battery or charging cable voltage to determine if the battery or charger is faulty.
Desktop PC suddenly shuts down	Measures the power supply (SMPS) output voltages (+3.3V, +5V, +12V) to verify the PSU is working correctly.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

- I watched a demonstration of a motherboard POST diagnostic card. The POST card displayed two-digit hexadecimal POST codes while the computer was starting up. The video showed that the boot process stopped at a specific code, which indicated a RAM (memory) problem. By checking the POST code against the motherboard's code list, the faulty hardware was quickly identified and replaced.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.

➤ **Example Tool:** HWMonitor

Steps performed:

1. Download and install HWMonitor (or Speccy / CrystalDiskInfo).
2. Run the application.
3. Capture a screenshot of the main dashboard showing:
 - CPU Temperature
 - RAM Information
 - Storage Health
 - Fan Speed (if available)
4. Blur or crop any personal information before submitting.

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Electrical voltage, current, and continuity	Faulty laptop charger or damaged power supply
POST Card	Motherboard POST (Power-On Self-Test) error codes	Defective RAM, CPU, or motherboard preventing boot
Software Diagnostics	Hardware health, temperatures, disk	Overheating CPU, failing hard

	status, and system information	drive/SSD, or abnormal temperatures
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